SRI CHAITANYA EDUCATIONAL INSTITUTIONS,INDIA.

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AIIMS GRAND TEST KEY

SR ELITE, SR AIIMS S60 & LTC-IC **DATE : 21-05-20**

PHYSICS

1) *c*	2) *c*	3) *a*	4) *c*	5) *d*	6) *d*	7) *a*	8) *c*	9) *c*	10) *d*
11) *a*	12) *d*	13) *d*	14) *b*	15) *a*	16) *b*	17) *b*	18) *d*	19) *b*	20) *d*
21) *c*	22) *d*	23) *c*	24) *a*	25) *a*	26) *a*	27) *b*	28) *c*	29) *a*	30) *a*
31) *a*	32) *a*	33) *b*	34) *c*	35) *c*	36) *b*	37) *c*	38) *c*	39) *b*	40) *c*
41) *a*	42) *b*	43) *a*	44) *a*	45) *b*	46) *a*	47) *a*	48) *b*	49) *a*	50) *a*
51) *a*	52) *a*	53) *c*	54) *c*	55) *c*	56) *a*	57) *b*	58) *a*	59) *b*	60) *b*

CHEMISTRY

61) *d*	62) *b*	63) *c*	64) *d*	65) *a*	66) *b*	67) *a*	68) *c*	69) *c*	70) *d*
71) *b*	72) *b*	73) *d*	74) *b*	75) *a*	76) *a*	77) *d*	78) *b*	79) *d*	80) *a*
81) *b*	82) *d*	83) *b*	84) *c*	85) *a*	86) *c*	87) *d*	88) *c*	89) *d*	90) *d*
91) *d*	92) *c*	93) *b*	94) *d*	95) *c*	96) *b*	97) *b*	98) *d*	99) *c*	100) *b*
101) *a*	102) *a*	103) *d*	104) *c*	105) *a*	106) *a*	107) *a*	108) *c*	109) *d*	110) *a*
111) *b*	112) *b*	113) *c*	114) *b*	115) *d*	116) *c*	117) *c*	118) *b*	119) *d*	120) *d*

BOTANY

121) *b*	122) *d*	123) *a*	124) *c*	125) *c*	126) *a*	127) *d*	128) *a*	129) *d*	130) *a*
131) *a*	132) *b*	133) *c*	134) *a*	135) *a*	136) *d*	137) *c*	138) *d*	139) *b*	140) *b*
141) *b*	142) *d*	143) *c*	144) *c*	145) *b*	146) *b*	147) *d*	148) *b*	149) *c*	150) *c*

ZOOLOGY

151) *d*	152) *c*	153) *c*	154) *c*	155) *c*	156) *b*	157) *d*	158) *b*	159) *a*	160) *b*
161) *b*	162) *d*	163) *b*	164) *b*	165) *a*	166) *b*	167) *c*	168) *b*	169) *c*	170) *a*
171) *d*	172) *b*	173) *c*	174) *b*	175) *a*	176) *c*	177) *c*	178) *b*	179) *b*	180) *b*

GENERAL KNOWLEDGE AND APTITUDE

181) *b*	182) *d*	183) *b*	184) *c*	185) *b*	186) *c*	187) *b*	188) *c*	189) *d*	190) *b*
191) *d*	192) *c*	193) *c*	194) *c*	195) *a*	196) *b*	197) *d*	198) *b*	199) *c*	200) *d*

SOLUTIONS

Physics:-

1. $\frac{1}{2}\times 4\times t^2 = 200+\frac{1}{2}\times 2\times t^2$

2. $a \propto \frac{1}{r}$

3. $\frac{l}{L}=\frac{\mu}{\mu+1}$

4. a = 6t

   $v = 3t^2$

   $u = 0, V = 3ms^{-1}$

   $W = \Delta K$

5. $\Delta P = P(1+e)+ep(1+e)+e^2p(1+e)+...$

   $= P\left(\frac{1+e}{1-e}\right)$

6. $20\times 10 = (80+20)s$

   $S = 2m$

   Distance moved by man = 10 – 2 = 8 m

   Distance form shore = 25 – 8 = 17 m

7. $T_1 + T_2 = 3mg$

   Taking torque about left end.

   $mg\times 50 + 2mg\times 75 = T_2\times 100$

   $T_2 = 2mg$

   $T_1 = mg$

8. $I = \frac{ml^2}{3}+\frac{ml^2}{3}+\left[\frac{ml^2}{12}+m\left(\frac{\sqrt{3}}{2}l\right)^2\right] = 3mk^2$

9. $T^2 \propto r^3$

10. $\frac{\Delta l}{l}=\frac{stress}{y}$ $\qquad$ $\frac{\Delta l}{V}=2\frac{\Delta r}{r}+\frac{\Delta l}{l}$

11. $mg - B = KV^2$

12. $\Delta T = T\left[\frac{1}{r_1}+\frac{1}{r_2}\right]$ here $r_2 = 0$

    $\Delta P = \frac{T}{r}$

13. $100\times 0.1\times(T-75)$

    $= 100\times 0.1\times 45 + 170\times 1\times 45$

14. Rouault expansion of hole

15. $\Delta\theta \propto R$

16. $W = \frac{nR}{\delta-1}(T_1 - T_2)$

17. $\eta = \frac{W}{Q_1}$

18. Free expansion

19. $KE = \frac{1}{2}mA^2\omega^2\cos^2\omega t$

20. $C_V = \frac{5R}{2}$ $\qquad$ $C_P = \frac{7R}{2}$

21. $PV = nRT$

22. Number of beats $\Delta n = \frac{2nx}{V}$

    Where *x* is the velocity of toy car.

23. Potential constant, hence $E \propto \frac{1}{r}$

24. $dv = -\bar{E}.d\bar{r}$

    $V_P - V_0 = -(25\bar{i}+30\bar{j}).(2\bar{i}+2\bar{j})$

25. $R \propto \frac{1}{r^4}$

26. Conceptual

27. $\bar{F} = q(\bar{V}\times\bar{B})$

28. Magnetization $= \frac{M}{V} = \frac{NIA}{A\times l} = \frac{NI}{l}$

29. Conceptual

30. $\varepsilon = -NA\frac{dB}{dt}$

31. $I_0 = \frac{\varepsilon_0}{L\omega}$

32. Propagation = $\bar{E}\times\bar{B}$

    And E = CB

33. Magnification $m = \frac{f_0}{f_e} = 30$

And $\theta_I = m.\theta_0$

Where $\theta_0 = \frac{50}{1000} = \frac{1}{20}$ radiations.

34. $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$

35. $d \sin\theta = n\lambda$

$d \sin\theta = \lambda$

$\frac{d}{a} = n$

36. $\lambda_n = n \times 2\pi r_0$

37. Conceptual

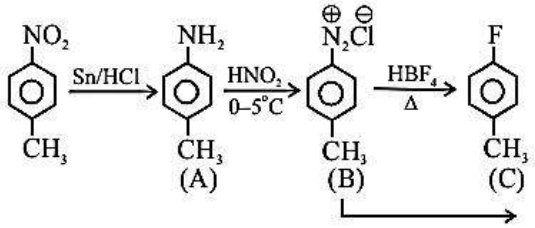
38. $N = N_0 e^{-\lambda t}$

Here $t = \frac{1}{\lambda}$

39. $\Delta\phi = 180^o$

40. Output is given by and gate. Hence its two inputs are to be 1 each

62.

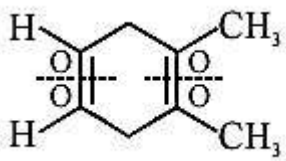


Balz-Schiemann reaction

63.

Due to back bonding in BF_3, B—F bond attains partial double bond character.

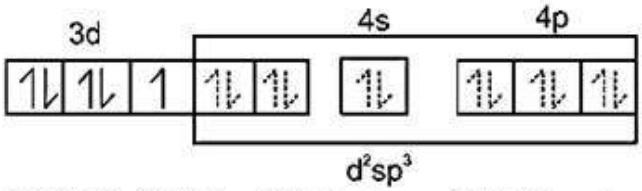
64.



67.

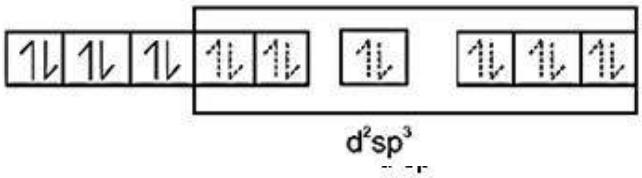
(1) $[FeCl(CN)_4(O_2)]^{+}$; O_2 is O_2^- ; CN^- is strong field ligand, so compels for the pairing of electrons. Compound is paramagnetic in nature due to superoxide ion

$Fe^{+2} \rightarrow 3d^6$;



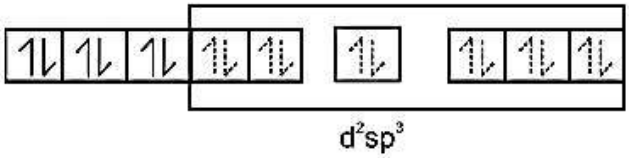
(2) $K_4[Fe(CN)_6]$; CN^- is strong field ligand so compels for the pairing of electrons.

$Fe^{2+} \rightarrow 3d^6$:



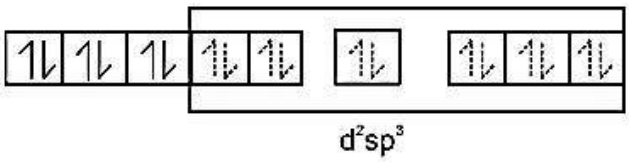
(3) $[Co(NH_3)_6]Cl_3$; NH_3 is strong field ligand and $3d^6$ configuration has higher CFSE compelling for the pairing of electrons.

$Co^{3+} \rightarrow 3d^6$:

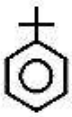


(4) $[Fe(CN)_5(O_2)]^{5-}$; O_2 is O_2^{2-} ; CN^- is strong field ligand so compels for the pairing of electrons.

$Fe^{2+} \rightarrow 3d^6$:



68



does not have benzylic H

$CH_3–\overset{\overset{O}{\|}}{C}–NH_2$ does not give iodoform rxn. because α–H is not released as $H^{\oplus}$

69.

$\overset{+}{M}\ \overset{-}{F} + XeF_4 \rightarrow \overset{+}{M}\ XeF_5^{\ominus}$

$\Rightarrow$ All IA fluorides are $\overline{F}$ donor

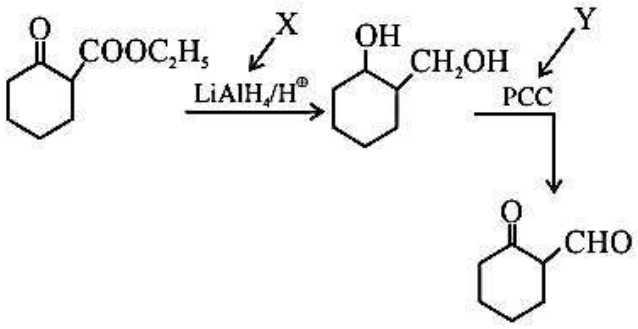
$\Rightarrow$ $XeF_5^{\ominus} \rightarrow sp^3d^3$

$\rightarrow$pentagonal planer.

71.

$(CH_3)_3C\bar{O}K^{\oplus}$ it is a bulky gp. so major product will be according to Hoffman's Rxn.

74.



$NaBH_4$ does not reduce Ester

H_2CrO_4 oxidise 1° alcohol to carboxylic acid

75.

$TeF_6 > SeF_6$

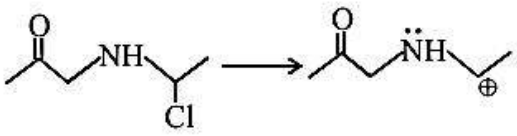
Due to large size of central atom.

CCl_4, PF_3 and NF_3 are not hydrolysed under normal condition.

77.

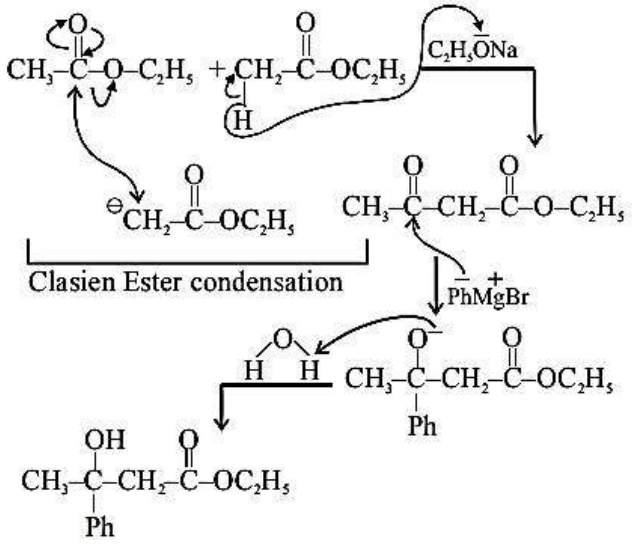
Reactivity towards α stability of $C^{\oplus}$

S_N1



Stable by resonance and 3α–H most stable among $C^{\oplus}$ formed from other halide

78.



79.

Acetic Acid exist in dimeric form in vapour state and in non-polar solvent.

80.

$$(C_v)_{molar} = \frac{R}{\gamma - 1}$$

$$(C_v)_{sp} = \frac{(C_v)_{molar}}{M}$$

82.

$E^{\circ}_{RP_{Cr_2O_7^{2-}/Cr^{3+}}} = 1.33V$ and $E^{\circ}_{RP_{Fe^{3+}/Fe^{2+}}} = 0.77V$

$\therefore$ $E^{\circ}_{OP_{Fe^{2+}/Fe}}$ is more thus it will oxidise or electron will flow from fe electrode to Cr electrode. Also, Fe electrode will be negative.

Also

$$E^{\circ}_{cell} = E^{\circ}_{OP_{Fe}} + E^{\circ}_{RP_{Cr}} = -0.77 + 1.33$$

83.

p and m-isomer form intermolecular H-bonding o-isomer form intramolecular H-bonding

84.

$$P.E. = \frac{kq_1q_2}{r}$$

$$P.E. = \frac{1}{4\pi\varepsilon_0}\frac{(-e)(+ze)}{r}$$

$z = 2$

85.

$$\lambda_{AgCl} = K \times \frac{1000}{M}$$

87. Only 1° amine give isocyanide test.

89.

An increase in rate of reaction in forward direction by a catalyst for a reaction in equilibrium brings in an increase in concentration of products and thus rate of backward reaction also increases to same magnitude and the equilibrium position is not altered.

90.

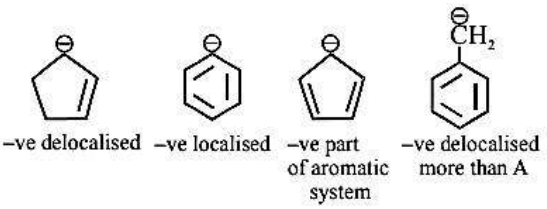
X-fcc = 4 Y-Tetrahedral void = 8

Z-1/2 of octahedral void = 2

Hence, $X_4Y_8Z_2 = X_2Y_4Z$

91.

Polarising power $\propto \phi$



92. So $\frac{\text{stability order}}{C > D > A > B}$

94.

(A) $NH_4NO_2 \xrightarrow{\Delta} N_2 + 2H_2O$,

(B) $NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$,

(C) $2NaN_3 \xrightarrow{\Delta} 3N_2 + 2Na$.

96.

$2P_y - 2P_y = \pi$ bond

$3P_y - 3P_y = \pi$ bond

$2P_x - 2P_y = \sigma$ bond

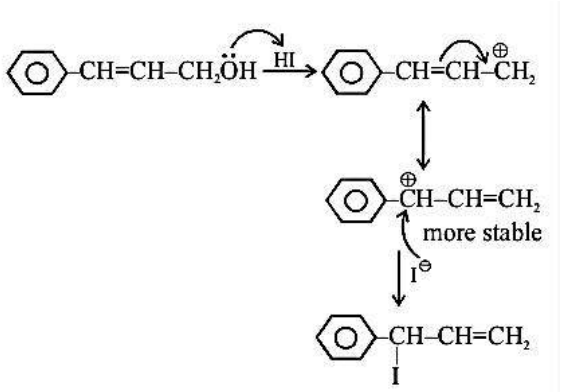
97.

$$BF_3 + 3LiBH_4 \longrightarrow 3LiF + 2B_2H_6$$

98.

XeF_4, XeF_5^-, XeF_3^+, ICl_4^-, $SCl_2 \rightarrow 2lp$

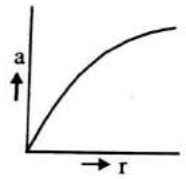
100.

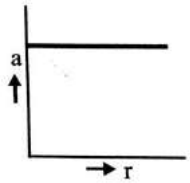


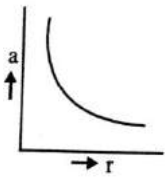
1. **A car is standing 200 m behind a bus, which is also at rest. The two start moving at the same instant but with different forward accelerations. The bus has acceleration $2m/s^2$ and the car has acceleration $4m/s^2$. The car will catch up with the bus after a time of**

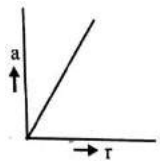
a) $\sqrt{110}s$ b) $\sqrt{120}s$ c) $10\sqrt{2}s$ d) $15s$

2. **If a body moving in circular path maintains constant speed of $10ms^{-1}$, then which of the following correctly describes relation between acceleration and radius ?**

a) 

b) 

c) 

d) 

3. **A uniform heavy chain is placed on a table with a part of it hanging over the edge. It just begins to slide when this part is one-third of its length. The coefficient of friction between the table and the chain is**

a) $\frac{1}{2}$ b) $\frac{1}{3}$ c) $\frac{2}{3}$ d) $\frac{3}{4}$

4. **A time dependent force F = 6t acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 second will be**

a) 9 J b) 18 J c) 4.5 J d) 22 J

5. **A ball falls vertically onto a floor, with momentum *p*, and then bounces repeatedly. The coefficient of restitution is *e*. The total momentum imparted by the ball to the floor is**

a) $p(1+e)$ b) $\frac{p}{1-e}$ c) $p\left(1+\frac{1}{e}\right)$ d) $p\left(\frac{1+e}{1-e}\right)$

6. **A boy of mass 20 kg is standing on a 80 kg free to move long cart. There is negligible friction between cart and ground. Initially, the boy is standing 25 m from a wall, If he walks 10 m on the cart towards the wall, then the final distance of the boy from the wall will be**

a) 15 m b) 12.5 m c) 15.5 m d) 17 m

7. **A uniform horizontal metre scale of mass *m* is suspended by two vertical string attached to its two ends. A body of mass *2m* is placed on the 75-cm mark. The tensions in the two strings are in the ratio**

a) 1 : 2 b) 1 : 3 c) 2 : 3 d) 3 : 4

8. **Three identical rods, each of length l, are joined to from a rigid equilateral triangle. Its radius of gyration about an axis passing through a corner and perpendicular to the plane of the triangle is**

a) $l/2$ b) $\sqrt{\frac{3}{2}}l$ c) $l/\sqrt{2}$ d) $l/\sqrt{3}$

9. The time period of a satellite of earth is 5 hours. If the separation between the earth and the satellite is increased to 4 times the previous value, the new time period will become

a) 10 hours b) 80 hours c) 40 hours d) 20 hours

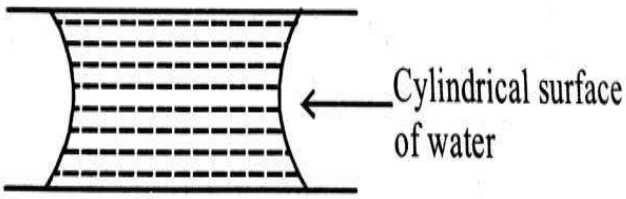
10. A uniform wire (Young's modulus $2\times10^{11}Nm^{-2}$) is subjected to longitudinal tensile stress of $5\times10^{7}Nm^{-2}$. If the overall volume change in the wire is 0.2%, the fractional decrease in the radius of the wire is close to :

a) 1.0×10^{-4} b) 1.5×10^{-4} c) 0.25×10^{-4} d) 8.75×10^{-4}

11. A spherical solid ball of volume V is made of a material of density ρ_1. It is falling through a liquid of density $\rho_2\left(\rho_2<\rho_1\right)$. Assume that the liquid applies a viscous force on the ball that is proportional to the square of its speed v, i.e., $F_{viscous}=-kv^2\left(k>0\right)$. The terminal speed of the ball is

a) $\sqrt{\frac{Vg\left(\rho_1-\rho_2\right)}{k}}$ b) $\frac{Vg\rho_1}{k}$ c) $\sqrt{\frac{Vg\rho_1}{k}}$ d) $\frac{Vg\left(\rho_1-\rho_2\right)}{k}$

12. Two glass plates have water between them and are separated by very small distance (see figure). If the radius of the cylindrical surface is R and surface tension of water is T then the excess pressure in water between the plates is :

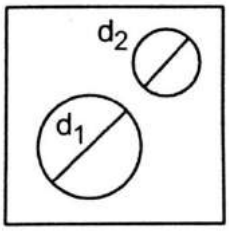


a) $\frac{2T}{R}$ b) $\frac{4T}{R}$ c) $\frac{T}{4R}$ d) $\frac{T}{R}$

13. A copper ball of mass 100 gm is at a temperature T. It is dropped in a copper calorimeter of mass 100 gm, filled with 170 gm of water at room temperature. Subsequently , the temperature of the system is found to be $75^{o}C$. T is given by (Given : room temperature = 30ºC, specific heat of copper = 0.1 cal/gmºC

a) 1250ºC b) 825ºC c) 800ºC d) 885ºC

14. Two holes of unequal diameters d_1 and $d_2\left(d_1>d_2\right)$ are cut in a metal sheet. If the sheet is heated,



a) Both d_1 and d_2 will decrease b) Both d_1 and d_2 will increase

c) d_1 will increase, d_2 will decrease d) d_1 will decrease, d_2 will increase

15. A large cylindrical rod of length L is made by joining two identical rods of copper and steel of length $\left[\frac{L}{2}\right]$ each. The rods are completely insulated from the surroundings. If the free end of copper rod is maintained at 100°C and that of steel at 0°C then the temperature of junction is (Thermal conductivity of copper is 9 times that of steel)

a) 90°C b) 50°C c) 10°C d) 67°C

16. During an adiabatic compression, 830 J of work is done on 2 mole of a diatomic ideal gas to reduce its volume by 50%. The change in its temperature is nearly: $\left(R = 8.3JK^{-1}mol^{-1}\right)$

a) 40 K b) 20 K c) 500 K d) 1000 K

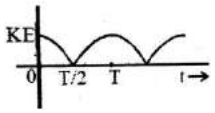
17. A Carnot engine absorbs 1000 J of heat energy from a reservoir at 127°C and rejects 600 J of heat energy during each cycle. The efficiency of engine is

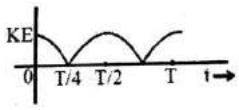
a) 20% b) 40% c) 50% d) 60%

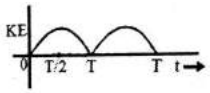
18. A container with insulating walls is divided into equal parts by a partition fitted with a valve. One part is filled with an ideal gas at a pressure P and temperature T, whereas the other part is completely evacuated. If the valve is suddenly opened, the pressure and temperature of the gas will be :

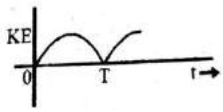
a) $\frac{P}{2},\frac{T}{2}$ b) P, T c) $P,\frac{T}{2}$ d) $\frac{P}{2},T$

19. A particle is executing simple harmonic motion with a time period T. At time t = 0, it is at its position of equilibrium. The kinetic energy-time graph of the particle will look like :

a) 

b) 

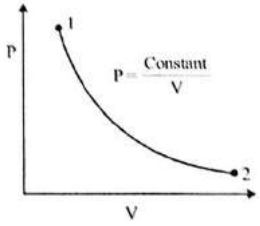
c) 

d) 

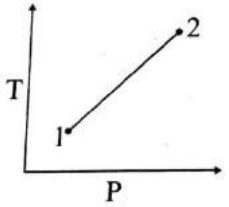
20. An ideal gas has molecules with 5 degree of freedom. The ratio of specific heats at constant pressure $\left(C_P\right)$ and at constant volume $\left(C_V\right)$ is :

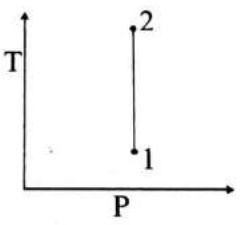
a) 6 b) $\frac{7}{2}$ c) $\frac{5}{2}$ d) $\frac{7}{5}$

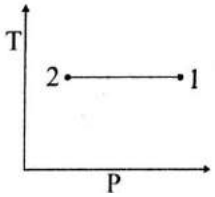
21. For the P-V diagram given for an ideal gas,

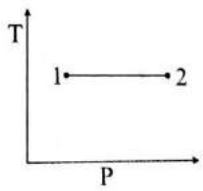


out of the following which one correctly represents the T-P diagram ?

a) 

b) 

c) 

d) 

22. A toy-car, blowing its horn, is moving with a steady speed of 5 m/s, away from a wall. An observer, towards whom the toy car is moving, is able to hear 5 beats per second . If the velocity of sound in air is 340 m/s, the frequency of the horn of the toy car is close to :

a) 680 Hz b) 510 Hz c) 340 Hz d) 170 Hz

23. Two spherical conductors A and B of radii 1 mm and 2 mm are separated by a distance of 5 cm and are uniformly charged. If the spheres are connected by a conducting wire then in equilibrium condition, the ratio of the magnitude of the electric fields at the surfaces of spheres A and B is

a) 4 : 1 b) 1 : 2 c) 2 : 1 d) 1 : 4

24. An electric field $\vec{E} = \left(25\hat{i} + 30\hat{j}\right) NC^{-1}$ exists in a region of space. If the potential at the origin is taken to be zero then the potential at $x = 2$ m, $y = 2$ is :

a) -110 V b) -140 V c) -120 V d) -130 V

25. A uniform wire of length l and radius r has a resistance of 100Ω. It is recast into a wire of radius $\frac{r}{2}$. The resistance of new wire will be :

a) 1600Ω b) 400Ω c) 200Ω d) 100Ω

26. In a sensitive meter bridge apparatus the bridge wire should possess

a) High resistivity and low temperature coefficient of resistivity
b) Low resistivity and high temperature coefficient of resistivity
c) Low resistivity and low temperature coefficient of resistivity
d) High resistivity and high temperature coefficient of resistivity

27. A negative test charge is moving near a long straight wire carrying a current. The force acting on the test charge is parallel to the direction of the current. The motion of the charge is :

a) Away from the wire b) Towards the wire
c) Parallel to the wire along the current d) Parallel to the wire opposite to the current

28. A 25 cm long solenoid has radius 2 cm and 500 total number of turns. It carries a current of 15A. If it is equivalent to a magnet of the same size and magneticzation $\vec{M}$ (magnetic moment/volume), then $\left|\vec{M}\right|$ is :

a) $30000\pi Am^{-1}$ b) $3\pi Am^{-1}$ c) $30000 Am^{-1}$ d) $300 Am^{-1}$

29. Curie temperature is the temperature above which

a) A ferromagnetic material becomes paramagnetic
b) A paramagnetic material becomes diamagnetic
c) A ferromagnetic material becomes diamagnetic
d) A paramagnetic material becomes ferromagnetic

30. A coil of circular cross-section having 1000 turns and $4cm^2$ face area is placed with its axis parallel to a magnetic field which decreases by $10^{-2} Wbm^{-2}$ in 0.01 s. The e.m.f. induced in the coil is :

a) 400 mV b) 200 mV c) 4 mV d) 0.4 mV

31. A sinusoidal voltage $V(t)=100\sin(500t)$ is applied across a pure inductance of L = 0.02 H. The peak current through the coil is :

a) 10A b) 5A c) 8A d) $5\sqrt{2}A$

32. Magnetic field in a plane electromagnetic wave is given by $\vec{B}=B_0\sin(kx+\omega t)\,\hat{j}T$ Expression for corresponding electric field will be :

a) $\vec{E}=B_0c\sin(kx+\omega t)\hat{k}V/m$ b) $\vec{E}=\frac{B_0}{c}\sin(kx+\omega t)\hat{k}V/m$

c) $\vec{E}=-B_0c\sin(kx+\omega t)\hat{k}V/m$ d) $\vec{E}=B_0c\sin(kx-\omega t)\hat{k}V/m$

33. A telescope, has an objective lens of focal length 150 cm and an eyepiece of focal length 5 cm. If a 50 m tall tower at a distance of 1 km is observed through this telescope in normal setting, the angle formed by the image of the tower is θ in radians is

a) 3 b) 1.5 c) 6 d) 2

34. You are asked to design a shaving mirror assuming that a person keeps it 10 cm from his face and views the magnified image of the face at the closest comfortable distance of 25 cm. The radius of curvature of the mirror would then be :

a) 60 cm b) -24 cm c) – 60 cm d) 24 cm

35. In a Young's double slit experiment, the distance between the two identical slits is 6 times larger than the slit width. Then the number of intensity maxima observed within the central maximum of the single slit diffraction pattern is :

a) 3 b) 6 c) 12 d) 24

36. The de-Broglie wavelength associated with the electron in the n = 4 level is :

a) $\frac{1}{4}$ th of the de-Broglie wavelength of the electron in the ground state.

b) Four times the de-Broglie wavelength of the electron in the ground state

c) Two times the de-Broglie wavelength of the electron in the ground state

d) Half of the de-Broglie wavelength of the electron in the ground state

37. In a radioactive series, $^{238}_{92}U$ changes to $^{206}_{82}Pb$ through $n_1\alpha$ -decay processes and $n_2\beta$ -decay processes.

a) $n_1=8,n_2=8$ b) $n_1=6,n_2=6$ c) $n_1=8,n_2=6$ d) $n_1=6,n_2=8$

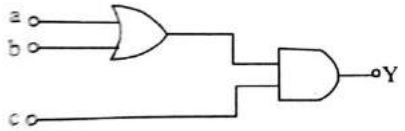
38. In a sample of radioactive material, what percentage of the initial number of active nuclei will decay during one mean life?

a) 37% b) 50% c) 63% d) 69.3%

39. In a common emitter amplifier circuit using an n-p-n transistor, the phase difference between the input and the output voltages will be :

a) 135º b) 180º c) 45º d) 90º

40. To get an output of 1 from the circuit shown in figure the input must be :



a) $a=0,b=0,c=1$ b) $a=1,b=0,c=0$

c) $a=1,b=0,c=1$ d) $a=0,b=1,c=0$

Directions for Qs. 41 to 60. In each of the following questions a statement of Assertion is given followed by a corresponding statement of reason just below it. Of the statements mark the correct answer as:

(a) If both assertion and reason are true and the reason is the correct explanation of assertion.

(b) If both assertion and reason are true but the reason is not the correct explanation of assertion

(c) If assertion is true statement but reason is false

(d) If both the assertion and reason are false

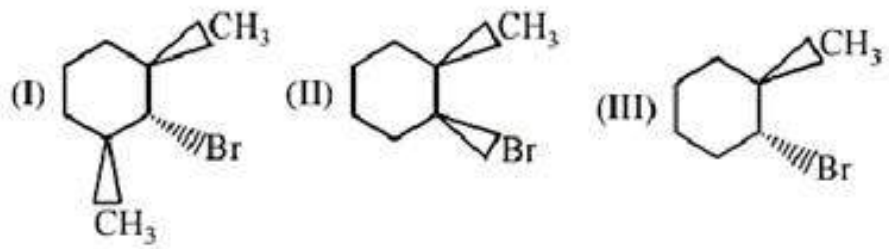
41. Assertion (A): A geostationary satellite must be located in the equitorial plane, that is at some point vertically above the equator.
    Reason (R): The only external force acting on an earth satellite is directed towards the centre of earth.
42. Assertion (A): When an uncharged, parallel plate capacitor is charged by connecting it to a cell, the heat produced in the circuit is equal to the energy stored in the capacitor
    Reason (R): The charge on a parallel plate capacitor means the equal and opposite charges on its inner surfaces.
43. Assertion (A): When a charged particle enters a magnetic field from outside, it cannot complete one rotation inside the field.
    Reason (R): The entry and exit of a charged particle into and out of a uniform magnetic field are symmetrical.
44. Assertion (A): It is possible for a charged particle to move in a circular path around a uniformly charged long conductor.
    Reason (R): The electrostatic force on the moving particle is directed towards the conductor.
45. Assertion (A): When two sounds of slightly different frequencies are heard together, beats are observed. Similar phenomenon is not observed when two lights of slightly different wavelengths reach a point together.
    Reason (R): Sound waves are longitudinal in nature, while light waves are transverse
46. Assertion (A): When a sphere falls under gravity or moves up due to buoyancy forces in a fluid, its velocity becomes constant after some time
    Reason (R): The force of viscosity is proportional to velocity
47. Assertion (A): It is necessary to define two molar heat capacities for a gas.
    Reason (R): Work is done by a gas when it volume changes
48. Assertion (A): The rate of conduction of heat through a rectangular slab is proportional to the difference in temperature between its end faces.
    Reason (R): The thermal conductivity coefficient of a rectangular slab of a material can be defined only under steady state conductions.
49. Assertion (A): In high quality optical devices like binocullors, prisms are used in place of plane mirrors to reflect light.
    Reason (R): In plane mirrors, reflection occur both in the front and rear (silvered) surfaces.
50. Assertion (A): When an electric heater is switched on, the colour of the filament changes from red to yellow to almost white.

Reason (R): Wien's displacement law states that $\lambda_m T = cons\tan t$

51. Assertion (A): When a coil is connected to cell, no current flows throught it initially.
    Reason (R): When a coil is connected to a cell, the initial emf induced in it is equal to the emf of cell.
52. Assertion (A): In a moving train, a small P,D arises across the axles of the wheels due to the earth's magnetic field. This P.D vanishes at the equator.
    Reason (R): At the equator, the earth's magnetic field is horizontal
53. Assertion (A): The number of active nuclei present in a radioactive sample decreases exponentially with time.
    Reason (R): The activity of a sample decreases linearly with time.
54. Assertion (A): Doppler effect occurs in all wave motions
    Reason (R): The Doppler effect can be explained by the principle of superposition of waves.
55. Assertion (A): A real image is formed when two rays, starting from a point, and moving along different paths, meet again at some other point.
    Reason (R): A real image can never be formed by reflection in a convex mirror or refraction in a concave lens.
56. Assertion (A): Any process which occurs very rapidly is likely to be adiabatic.
    Reason (R): Exchange of heat between a system and its surroundings is not instantanious.
57. Assertion (A): An open organ pipe can be used as a musical instrument but not a closed organ pipe.
    Reason (R): The fundamental frequency of an open pipe is twice the fundamental frequency of a closed organ pipe of the same length.
58. Assertion (A): Electric and magnetic lines of force do not intersect.
    Reason (R): The tangent drawn to a line of force gives the direction of the field at that point.
59. Assertion (A): The wavelength of K_α x-rays is higher than K_β x-rays for the same element
    Reason (R): Characteristic x-rays are due to orbital transition of electrons.
60. Assertion (A): In an elastic collision between two bodies the relative velocities before and after collision are equal.
    Reason (R): Linear momentum of system is conserved in elastic collisions.

CHEMISTRY

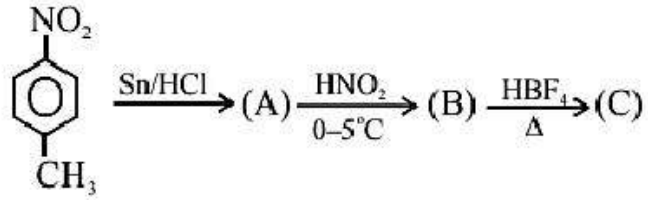
61.



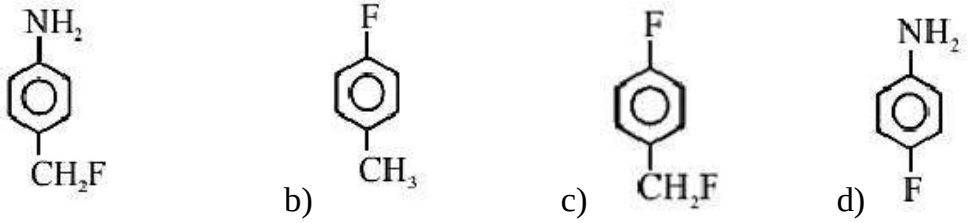
Ease of β-dehydrobromination

a) I > II > III b) III > II > I c) II > I > III d) II > III > I

62.



What ic C

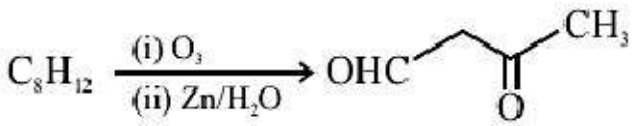


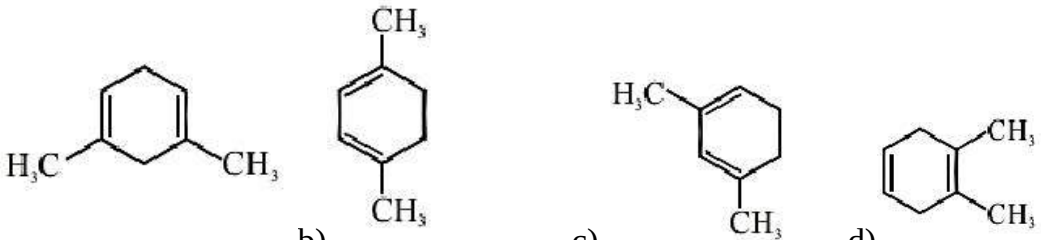
a) b) c) d)

63. In which of the following compounds B – F bond length is shortest?

a) BF_4^- b) $BF_3 \leftarrow NH_3$ c) BF_3 d) $BF_3 \leftarrow N\left(CH_3\right)_3$

64. What will be the structure of C_8H_{12}





a) b) c) d)

65. $K_2Cr_2O_7$ **is preferred to** $Na_2Cr_2O_7$ **for use in volumetric analysis as a primary standard because**

a) $Na_2Cr_2O_7$ is hygroscopic while $K_2Cr_2O_7$ is not

b) $K_2Cr_2O_7$ is hygroscopic while $Na_2Cr_2O_7$ is not

c) $K_2Cr_2O_7$ is pure while $Na_2Cr_2O_7$ is impure

d) None of these

66. Decimolar solution of potassium ferricyanide, $K_3\left[Fe(CN)_6\right]$ **has osmotic pressure of 3.94 atm at** 27^0C**. Hence percent ionization of the solute is**

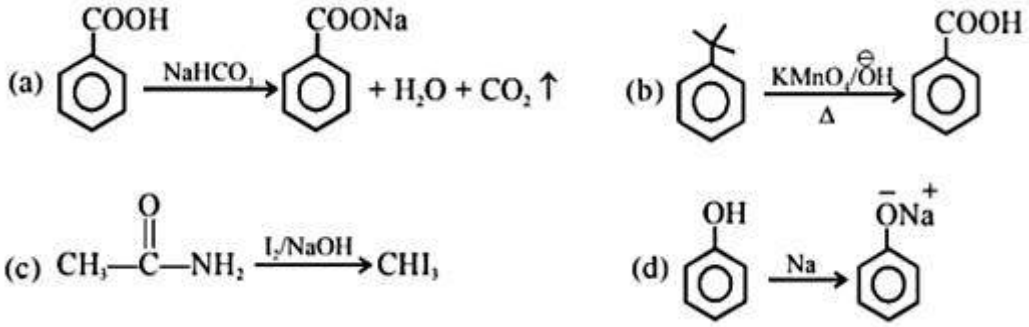
a) 10% b) 20% c) 30% d) 40%

67. All the metal ions contains $t_{2g}^{\ 6}e_g^0$ **configurations. Which of the following complex will be paramagnetic?**

a) $\left[FeCl(CN)_4\left(O_2\right)\right]^{4-}$ b) $K_4\left[Fe(CN)_6\right]$

c) $\left[Co\left(NH_3\right)_6\right]Cl_3$ d) $\left[Fe(CN)_5\left(O_2\right)\right]^{-5}$

68. Which of the following reaction not feasible



a) only b b) b, c, d c) b, c d) only c

69. $MF + XeF_4 \rightarrow A$ **(M^+ = Alkali metal cation). The state of hybridization of the central atom in 'A' and shape of the species are**

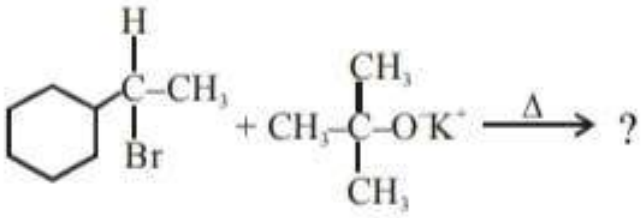
a) sp^3d , TBP b) sp^3d^2 (Distorted octahedral)

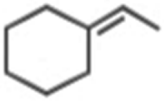
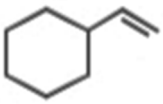
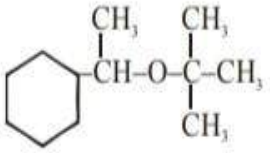
c) sp^3d^3 , pentagonal planer d) No compound formed at all

70. Which of the following metal shows self reduction in metallurgy process

a) Cu b) Hg c) Ag d) Both a and b

71. What is the major product of the reaction



a)  b)  c) 

d) Both a and c

72. Calculate the no. of moles of $Fe(OH)_3$ that can be produced by allowing 1mol of Fe_2S_3, 2mol of H_2O and 3 mol of O_2 to react according to following reaction

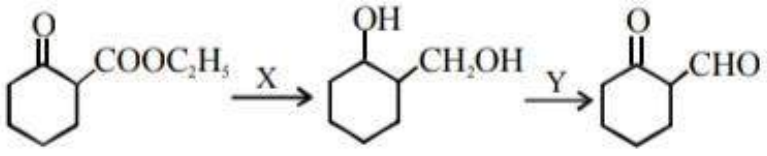
$2Fe_2S_3(s) + 6H_2O(l) + 3O_2(g) \rightarrow 4Fe(OH)_3(s) + 6S(s)$

a) 4 b) 1.33 c) 2 d) 0.66

73. The element that shows greater ability to form p–p multiple bonds, is :

a) So b) Ge c) Sn d) C

74.



The reagent X and Y can be

a) $NaBH_4 / H^{\oplus}$,PCC b) $LiAlH_4 / H^{\oplus}$, PCC

c) $LiAlH_4 / H^{\oplus}, H_2CrO_4$ d) $NaBH_4 / H^{\oplus}, H_2CrO_4$

75. Select correct order of ease of hydrolysis

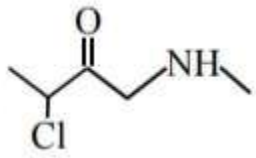
a) $TeF_6 > SeF_6$ b) $CCl_4 > SiCl_4$ c) $PF_3 > PCl_3$ d) $NF_3 > NCl_3$

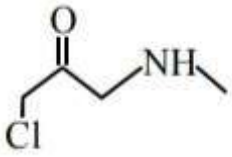
76. The quantity $\frac{PV}{k_BT}$ represents the (k_B : Boltzmann constant)

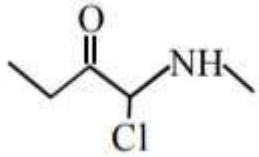
a) number of particles of the gas b) mass of the gas

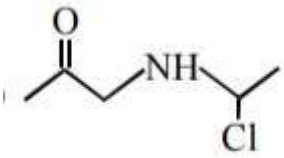
c) number of moles of the gas d) translation energy of the gas

77. Which is maximum reactive for S_N1 reaction of halide

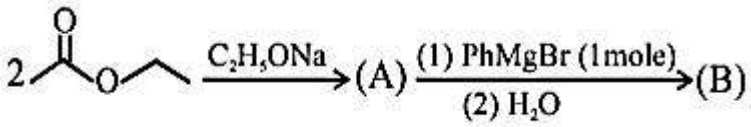
a) 

b) 

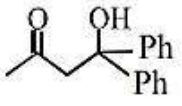
c) 

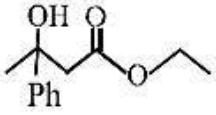
d) 

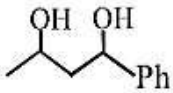
78.

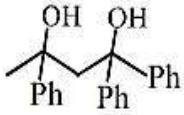


Then B is

a) 

b) 

c) 

d) 

79. Consider the following statements

I. HCl is more volatile than HF

II. In benzene, acetic acid dimerises

III. CH_3OCH_3 is non – polar but soluble in water.

Select the correct set of codes using T for true and F for false statements in given sequence

a) F T F b) F T F c) T F T d) T T F

80. For an ideal gas $\frac{C_{p,m}}{C_{v,m}} = \gamma$. The molecular mass of the gas is M, its specific heat capacity at constant volume is

a) $\frac{R}{M(\gamma-1)}$ b) $\frac{M}{R(\gamma-1)}$ c) $\frac{\gamma RM}{\gamma-1}$ d) $\frac{\gamma R}{M(\gamma-1)}$

81. The vander wall's equation for 0.5 mol gas is?

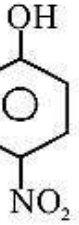
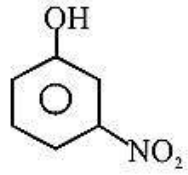
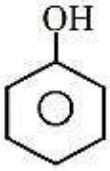
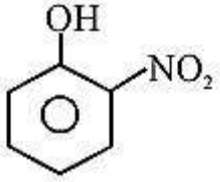
a) $\left(p+\frac{a}{4v^2}\right)\left(\frac{v-b}{2}\right)=\frac{2RT}{2}$ b) $\left(p+\frac{a}{4v^2}\right)(2v-b)=RT$

c) $\left(p+\frac{a}{4v^2}\right)(2v-4b)=RT$ d) $\left(p+\frac{a}{4v^2}\right)=\frac{2RT}{2(v-b)}$

82. For the cell prepared from electrode A and B; Electrode A : $Cr_2O_7^{2-}\,|\,Cr^{3+}, E^o_{red}=+1.33V$ **and Electrode B :** $Fe^{3+}\,|\,Fe^{2+}, E^o_{red}=0.77V$ **. Which of the following statement is correct?**

a) The electrons will flow from B to A when connection are made
b) The standard emf of the cell will be 0.56 V
c) A will be positive electrode
d) All of these

83. The correct order of boiling point is

(A) 
(B) 
(C) 
(D) 

a) C > A > D > B b) A > D > B > C c) D > A > B > C d) A > B > D > C

84. Potential energy of electron present in He^+ **is**

a) $\frac{e^2}{2\pi\varepsilon_0 r}$ b) $\frac{3e^2}{4\pi\varepsilon_0 r}$ c) $\frac{-2e^2}{4\pi\varepsilon_0 r}$ d) $\frac{-e^2}{4\pi\varepsilon_0 r^2}$

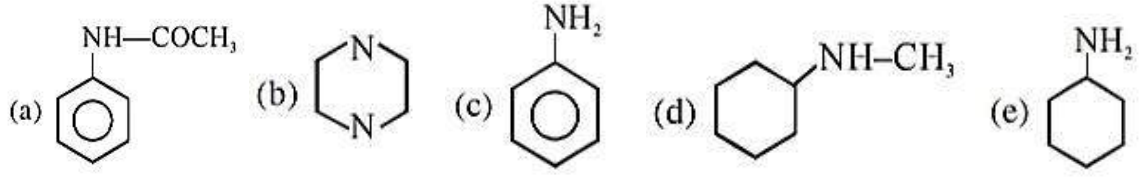
85. The specific conductivity of a saturated solution of AgCl is $3.40\times10^{-6}\Omega^{-1}cm^{-1}$ **at** 25^0C **. If** $\lambda_{Ag}=62.3\Omega^{-1}cm^2mol^{-1}$ **and** $\lambda_{Cl^-}=67.7\Omega^{-1}cm^2mol^{-1}$**, the solubility of AgCl at** 25^0C **is:**

a) $2.6\times10^{-5}\,\text{molL}^{-1}$ b) $3.731\times10^{-3}\,\text{mol L}^{-1}$
c) $3.731\times10^{-5}\,\text{mol L}^{-1}$ d) $2.6\times10^{-3}\,\text{g L}^{-1}$

86. Colloidal solution of silver is prepared by

a) Dialysis b) Double decomposition method
c) Bredig's electric arc method d) Peptization

87. Which among the following give positive isocyanide test



a) a, d b) only b c) only c d) c and e

88. A cyclic skeleton of silicon and oxygen can be constructed by the silicate ion with composition

a) $Si_2O_7^{-4}$ b) $Si_2O_5^{-2}$ c) SiO_3^{-2} d) SiO_4^{-4}

89. In a reversible reaction at equilibrium, a catalyst

a) Increases the rate of the forward reaction only

b) Increases the rate of the forward reaction to a greater extent than that of the backward reaction

c) Increases the rate of the forward reaction and decreases that of the backward reaction to different extent

d) Increases the rate of the forward and backward reactions simultaneously with equal extent

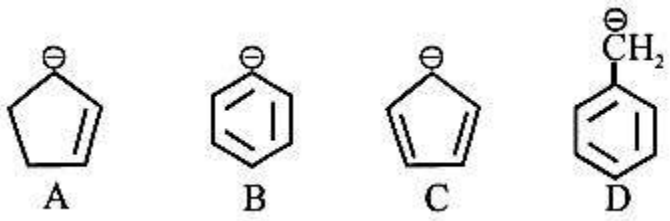
90. A solid is formed and it has three types of atoms X, Y and Z. X forms a fcc lattice with Y atoms occupying all tetrahedral voids and Z atoms occupying half of octahedral voids. The formula of solid is

a) X_4YZ_2 b) X_4Y_2Z c) XY_2Z_4 d) X_2Y_4Z

91. Which of the following polarizing power order is not correct

a) $Ca^{+2} < Pb^{+2}$ b) $Na^{+} < Cu^{+}$ c) $Sn^{+2} < Sn^{+4}$ d) $Zn^{+2} > Cd^{+2}$

92. Arrange the following carbanions in correct order of stability



a) C > D > B > A b) D > C > A > B c) C > D > A > B d) D > C > B > A

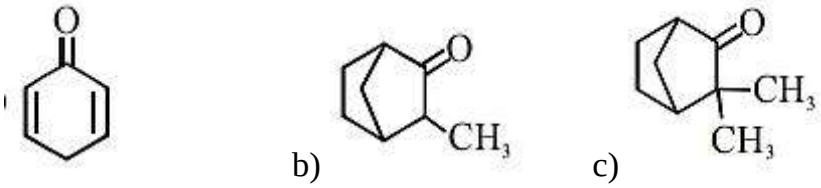
93. Decomposition of A follows first order kinetics by the following equation $4A(g) \rightarrow B(g) + 2C(g)$. If initially, total pressure was 800mm of Hg and after 10 minutes it is found to be 650 mm of Hg. What is half life of A? (Assume only A is present initially)

a) 10 min b) 5 min c) 7.5 min d) 15 min

94. Which of the following compounds does give N_2 on heating

a) NH_4NO_2 b) NH_4NO_3 c) NaN_3 d) a and c

95. Which of the following does not show tautomerism



a) b) c) d) All of these

96. The group of molecules having identical shape is

a) SF_4, XeF_4, CCl_4 b) ClF_3, $XeOF_2$, XeF_3^{+}

c) BF_3, PCl_3, XeO_3 d) PCl_5, IF_5, XeO_2F_2

97. In reaction, $BF_3 + 3LiBH_4 \rightarrow 3LiF + X$; here X is

a) B_4H_{10} b) B_2H_6 c) BH_3 d) B_3H_8

98. In Ramsay and Rayleigh's isolation of noble gases from air, the nitrogen of the air is finally converted into

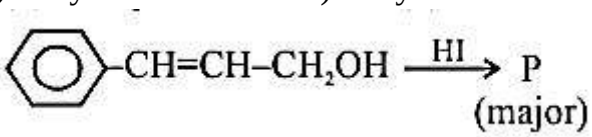
a) $NaNO_2$ only b) NO and NO_2

c) $NaNO_3$ only d) $NaNO_2$ and $NaNO_3$

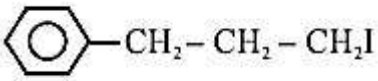
99. Solubility of AgCl in 0.2M NaCl is x and that in 0.1M $AgNO_3$ is y then which of the following is correct?

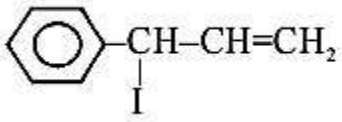
a) x = y b) x > y c) x < y d) we can not predict

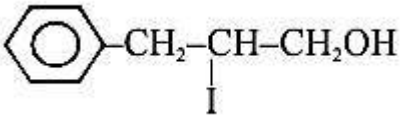
100.

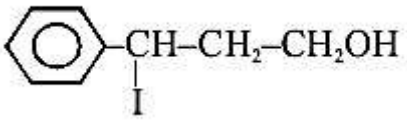


Product P is

a) 

b) 

c) 

d) 

Directions for Qs. 101 to 120. In each of the following questions a statement of Assertion is given followed by a corresponding statement of reason just below it. Of the statements mark the correct answer as:

(a) If both assertion and reason are true and the reason is the correct explanation of assertion.

(b) If both assertion and reason are true but the reason is not the correct explanation of assertion

(c) If assertion is true statement but reason is false

(d) If both the assertion and reason are false

101. Assertion (A): Alkali metals do not occur in native state in nature
Reason (R): Alkali metals are highly reactive metals

102. Assertion (A): Salicylic acid (o-hydroxy benzoic acid) is stronger acid than m and p siomers
Reason (R): Carboxylate ion is stablised by intra-molecular H-bonding.

103. Assertion (A): All the oxides of boron family with the general formula M_2O_3 are basic
Reason (R): From B_2O_3 to Tl_2O_3 basic character decreases

104. Assertion (A): Biphenyl is an Aromatic hydrocarbon
Reason (R): All Aromatic compound contain Benzene ring

105. Assertion (A): ΔS , ΔG and ΔH are zero for cyclic process
Reason (R): Free energy, entropy and enthalpy are state functions

106. Assertion (A): Hydrogen attached to SP^3 carbon in cyclopentadiene can be released as proton
Reason (R): Conjugate base of cyclopentadiene is aromatic

107. Assertion (A): Orbital angular momentum of 1s, 2s, 3s, etc is zero
Reason (R): Value of l for s – suborbit is zero

108. Assertion (A): HNO_2 may act as an oxidizing as well as a reducing agent
Reason (R): The oxidation number of Nitrogen remains same in all the compounds

109. Assertion (A): A non volatile solute is mixed in a solvent then elevation in boiling point and depression of freezing point both are equal

Reason (R): Elevation in boiling point and depression in freezing point both depend on melting point of non volatile solute

110. Assertion (A): $CH_3CH_2OH + NaCl + H_2SO_4 \rightarrow CH_3CH_2Cl$ Above reaction does not take place in absence of H_2SO_4

Reason (R): $\overset{\Theta}{O}H$ is poor leaving group

111. Assertion (A): The value of Δ_0 for M^{3+} complexes are always much higher than value for M^{2+} complexes (for the same set of ligands)

Reason (R): The crystal field stabilization energy of $\left[Co\left(NH_3\right)_6\right]^{3+} < \left[Rh\left(NH_3\right)_6\right]^{3+}$

112. Assertion (A): Protonation of the carbonyl group increases it's electrophilic nature

Reason (R): This is an example of electrophilic addition on nucleophilic oxygen

113. Assertion (A): CH_3OH and CH_3CH_2OH can be distinguished by haloform test

Reason (R): Haloform test is given by all 2^0 alcohol

114. Assertion (A): Ionisation energy of Zn is higher than Ga

Reason (R): Zn has d^{10} configuration in its penultimate shell

115. Assertion (A): On heating ammonium dichromate, NH_3 gas is produced

Reason (R): $Cr_2O_7^{2-}$ ion is a strong Reducing agent

116. Assertion (A): Nucleophilicity of $\overset{\Theta}{N}H_2$ is greater than $\overset{\Theta}{O}H$

Reason (R): Basic nature of $\overset{\Theta}{O}H$ is higher than $\overset{\Theta}{N}H_2$

117. Assertion (A): In any ionic solid [MX] with Schottky defects, the number of positive and negative ions are same

Reason (R): Equal number of cations and anions are present at interstitial space

118. Assertion (A): The formula of azurite is $2CuCO_3.Cu(OH)_2$

Reason (R): The composition of malachite ore $CuCO_3.Cu(OH)_2$

119. Assertion (A): The negative sign of reduction potential indicates that an electrode when joined with SHE acts as cathode

Reason (R): Substances with negative reduction potential are good oxidising agents

120. Assertion (A): Energy of the orbitals in multi electron system increases as 1s < 2s = 2p < 3s = 3p = 3d < 4s = 4p = 4d = 4f <….

Reason (R): Energy of the electron depends completely on principal quantum number

BOTANY

121. Match the following columns and select the correct option

Column-I	Column-II
A) *Triticum aestivum*	**1) Mango**
B) *Mangifera indica*	**2) Cat**
C) *Musca domestica*	**3) Dog**

D) *Canis lupus familiaris*	**4) Wheat**
E) *Felis catus*	**5) Housefly**

Codes :

	A	B	C	D	E
a)	2	5	1	3	4
b)	4	1	5	3	2
c)	2	5	3	1	4
d)	4	1	5	2	3

122. Which of the following is a 'cyanophage' ?

a) S-13 b) $\phi \times 174$ c) SV-40 d) LPP-1

123. Shiitake mushroom is

a) *Lentinus edodes* b) *Volvoriella volvaceae*

c) *Agaricus bisporus* d) *Plerotus sojar-Caju*

124. Dikaryotisation (n+n) in *Agaricus* is brought about by

a) clamp connections

b) somatogamy between two hyphae of different strains

c) both a and b

d) basidiospores

125. Which of the following helps in respiration of lichens ?

a) Isidia b) Soredia c) Cyphella d) Cephalodia

126. 'Cladode' is a characteristic morphological feature of

a) *Asparagus and Ruscus* b) *Casuarina and Opuntia*

c) *Cladophora and Cactus* d) *Citrus and Euphorbia*

127. In the sieve elements, which one of the following is most likely the function of P-proteins?

a) Deposition of callose on sieve plates b) Providing energy for active translocation

c) Autolytic enzymes d) Sealing mechanism on wounding

128. Satellite of chromosome is

a) rich in RNA and deficient in DNA b) rich in DNA and deficient in RNA

c) rich in protein d) lacks DNA

129. Cellulose, the most important constituent of plant cell wall is made up of

a) branched chain of glucose molecules linked by α, 1-6 glycosidic bond at the site of branching

b) unbranched chain of glucose molecules linked by α, 1-4 glycosidic bond

c) branched chain of glucose molecules linked by β, 1-4 glycosidic bond in straight chain and α, 1-6 glycosidic bond at the site of branching

d) unbranched chain of glucose molecules linked by β, 1-4 glycosidic bond

130. Spindle fibres of mitotic cells are made up of

a) tubulin b) actin c) myosin d) collagen

131. In rainy season, doors get swelled due to

a) imbibition b) diffusion

c) transpiration d) respiration

132. Leghaemoglobin helps in

a) nitrogen-fixation b) protecting nitrogenase from O_2

c) destroy bacteria d) transport of food in plants

133. The total requirement of ATP and NADPH for each molecule of CO_2 fixed and reduced in photosynthesis in the Calvin cycle is

a) 2 ATP and 2 NADPH b) 2 ATP and 3 NADPH

c) 3 ATP and 2 NADPH d) 4 ATP and 3 NADPH

134. How many types of cytochromes are involved in respiratory electron transport chain ?

a) Five b) Two c) Six d) Four

135. The sequence of development during the formation of embryosac is

a) Archesporium → Megaspore mother cell → Megaspore → Embryo sac

b) Archesporium → Megaspore → Megaspore mother cell → Embryo sac

c) Megaspore → Archesporium → Megaspore mother cell → Embryo sac

d) Megaspore mother cell → Spore mother cell → Embryo spore mother cell → Embryo sac

136. Which one of the following best describes polygenic inheritance ?

a) ABO blood group in human and flower colour of *Mirabilis jalapa*

b) Hair pigment of mouse and tongue rolling in humans

c) Human eye colour and sickle-cell anaemia

d) Human height and colour of skin and eyes

137. In the *lac* operon model, lactose molecules function as

a) inducers, which bind with the operator gene

b) represors, which bind with the operator gene

c) inducers, which bind with the repressor protein

d) corepressors, which bind with repressor protein

138. Rice wine is

a) beer b) wine c) cider d) sake

139. This method of finding a gene is used when researchers know very little about the gene they are trying to find. This process results in a complete gene library a collection of copies of DNA fragments that represents the entire genome of an organism.

a) Cloning b) Shotgun cloning c) Gene synthesis cloning d) PCR

140. Grafting is successful in dicots but not in monocots because the dicots have

a) vascular bundles arranged in a ring

b) cambium for secondary growth

c) vessels with elements arranged end to end

d) cork cambium

Directions for Qs. 141 to 150 : In each of the following questions a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements mark the correct answer as:

(a) If both assertion and reason are true and the reason is the correct explanation of assertion.

(b) If both assertion and reason are true but the reason is not the correct explanation of assertion

(c) If assertion is true statement but reason is false

(d) If both the assertion and reason are false

141. Assertion (A): Recognition site should be preferably single and responsive to commonly used restriction enzyme.

Reason (R): In pBR322, alien DNA is ligated generally in the area of Bam HI site of tetracycline resistance gene.

142. Assertion (A): *Agrobacterium tumefaciens* is a popular genetic engineer because this bacterium is associated with the roots of all cereal and pulse crops.
Reason (R): A gene incorporated in the bacterial chromosomal genome gets automatically suppressed into the crop with which the bacterium is associated.

143. Assertion (A): Azolla pinnata is used as a biofertiliser in rice cultivation.
Reason (R): Azolla performs nitrogen fixation with the help of symbiotic bacterium Bacillus sp.

144. Assertion (A): Virus-free plants can be produced from virus infected plants by means of meristem tissue culture.
Reason (R): The virus fails to grow during the growth of host tissue in the artificial medium.

145. Assertion (A): Amber codon is a termination codon.
Reason (R): If in an mRNA, a termination codon is present, the protein synthesis stops abruptly whether the protein synthesis is complete or not.

146. Assertion (A): Coconut tree is grown in coastal areas over a large part of the world.
Reason (R): Coconut fruit can float and get dispersed over thousands of kilometer before losing viability.

147. Assertion (A): In heminatropous ovule, the funicle lies parallel to body or ovule.
Reason (R): In heminatropous ovule, body of ovule has rotated by 45°.

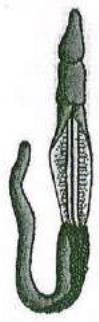
148. Assertion (A): Photomodulation of flowering is a phytochrome regulated process.
Reason (R): Phytochrome is a chromoprotein

149. Assertion (A): In TCA cycle, isomerisation of citric acid into isocitric acid is accompanied by the change in $CO_2\left(COO^-\right)$ position.
Reason (R): Isocitric acid is more stable than citric acid.

150. Assertion (A): F_1-particles are present in the inner mitochondrial membrane.
Reason (R): An electron gradient formed on the inner mitochondrial membrane forms ATP.

ZOOLOGY

151. Which statements are correct about the given animal?



(a) Bilaterally symmetrical, triploblastic **(b) Worm like fresh water animal**
(c) Circulatory system is of open type **(d) Coelomate animals**
(e) Fertilisation is internal, direct development

(a) a,b,c,d,e are correct (b) a,b,c,d are correct
(c) a,b,d are correct (d) a,c,d are correct

152. The nerve cord of cockroach is

(a) Single, dorsal & solid (b) Double, dorsal & hollow
(c) Double, ventral & solid (d) Single, ventral & hollow

153. Which is the incorrect statement for areolar connective tissue?

(a) Present below the skin
(b) It contains fibroblasts.
(c) Fibers are densely packed
(d) Works as support framework for epithelium

154. Estimate the number of species of tortoises on an island having an area of 3600 square kilometers, if the regression coefficient of the species-area relationship curve is 0.5 and Y-intercept is 0.1 –

(a) 9 (b) 3 (c) 6 (d) 12

155. Find out the incorrect one-

(a) hCG, hPL and relaxin are produced in women only during pregnancy.
(b) During pregnancy the levels of hormones like estrogens increased several folds.
(c) Foetal ejection reflex related with lactation.
(d) Foetal ejection reflex is related with parturition

156. Diaphragms are contraceptive devices used by the females. Choose the correct option from the statements given below:-

(a) They are introduced into the uterus
(b) They are placed to cover the cervical region
(c) They act as physical barrier for sperm entry
(d) They mainly function as spermicidal agents

The correct options are :-

(a) a & b (b) b & c (c) a & c (d) c & d

157. The karyotype of human ovum is n+1 due to nondisjunction of 21st pair of chromosomes. If, it is fertilized by a normal sperm with 'x' chromosome, it develops into

(a) Turner's female
(b) Klinefelter syndrome male
(c) Down syndrome-male
(d) Down syndrome – female

158. Origin of different varieties of finches from seed-eating finches explains

(a) Convergent evolution (b) Adaptive radiation
(c) Directional selection (d) Parallel evolution

159. The study of fossils in different aged rocks showed that most fossils exist in-

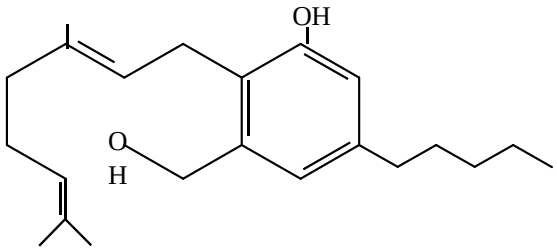
(a) Sedimentary rock (b) Igneous rock
(c) Marble (d) All of these

160. The immediate adverse effects of drugs and alcohol abuse are manifested in the form of

(a) Anxiety, shakiness, nausea, sweating
(b) Reckless behavior, vandalism, violence

(c) Death
(d) Fever

161. Which of the components of following options correctly applies to structural formula shown in the diagram?



(a)	Opium	Unripe fruit	Analgesic action
(b)	Hemp	Inflorescence	Effects cardiovascular system
(c)	Coca	Dopamine inhibition	CNS stimulation
(d)	Datura	Atropine	Hallu Cinations

162. A biologist studies the population of rats in a barn. He found that the average natality was 250, average mortality 240, Immigration 20 and Emigration 30. The net increase in population is-

(a) 10 (b) 15 (c) 05 (d) Zero

163. According to David Tilman, greater the diversity, greater is the primary productivity. Which of the following has least diversity but high primary productivity-

(a) Desert ecosystem (b) Cropland ecosystem
(c) Tropical rainforest (d) Composite fish culture pond.

164. Accumulation of toxic substance in successive trophic level in food chain is known as

(a) Accelerated eutrophication (b) Biomagnification
(c) Biotype (d) Aggregation

165. According to CPCB, particulate dust of size ----------in diameter, cause the greatest harm to human health-

(a) 2.5 micrometers or less (b) 2.5 micrometers or more
(c) 3.5 micrometers (d) 4.5 micrometers.

166. Secondary productivity is rate of formation of new organic matter by-

(a) Parasite (b) Consumer (c) Decomposer (d) Producer.

167. Which of the following genotype shows Intersex drosophila (as per genic balance theory)-

(a) 2A+XXX (b) 2A+XXY (c) 3A+XXY (d) 2A+XY

168. A Y-linked gene is responsible for hypertrichosis (Long hair on ears) . When an affected man marries a normal woman, what % of their daughters, would be expected to have hairy ears-

(a) 25 % (b) 0 % (c) 50 % (d) 100 %

169. DNA fingerprinting profile will be exactly same of-

(a) Non Identical twins (b) Di-zygotic twins
(c) Mono-zygotic twins (d) Siamase twins.

170. The probability of severe sickle cell disease when both parents are carriers-

(a) 1/4 (b) 1/2 (c) 1/3 (d) 3/4

Directions for Qs. 171 to 180. In each of the following questions a statement of Assertion is given followed by a corresponding statement of reason just below it. Of the statements mark the correct answer as:

(a) If both assertion and reason are true and the reason is the correct explanation of assertion.

(b) If both assertion and reason are true but the reason is not the correct explanation of assertion

(c) If assertion is true statement but reason is false

(d) If both the assertion and reason are false

171. Assertion (A): Maximum movement is possible at the amphiarthroses joint.
Reason (R): Such joints are also called synovial joints and have almost frictionless movement due to synovial fluid.

172. Assertion (A): Inulin is used in testing kidney function specially glomerular filtration.
Reason (R): Inulin is a fructan storage polysaccharide.

173. Assertion (A): Receptors in the tendon, joints give information regarding the position and movements of different parts of the body.
Reason (R): These are termed as Noci-receptors.

174. Assertion (A): Sharpest vision is in foves centralis.
Reason (R): The relationship of receptor to bipolar cells to ganglion cells is 1-1-1 within fovea centralis.

175. Assertion (A): Occasionally mammary glands are functional in males and the condition is called gynaecomastia.
Reason (R): Decreased testosterone in later life may also lead to gynaecomastia. It is usually due to perturbation of estrogen to androgen ratio.

176. Assertion (A): Oxytocin stimulates contraction of uterine muscles during birth and initiates ejection of milk.
Reason (R): It is synthesized in the posterior lobe of pituitary.

177. Assertion (A): Aptyalism patients have higher than normal incidences of dental carries.
Reason (R): Aptyalism is caused by the action of parasympathetic nervous system.

178. Assertion (A): Emphysema is a chronic obstructive disease of lung, causing irreversible distention and loss of elasticity of alveoli.
Reason (R): Emphysema is preventable if chronic exposure to smoke (cigarette and others) and pollutant is avoided.

179. Assertion (A): Latent period is the interval between the application of appropriate stimulus and initiation of contraction.
Reason (R): Latent period is minimum in cardiac muscle fibers.

180. Assertion (A): Fish meal is a rich source of protein for cattle and poultry.

Reason (R): Fish meal is prepared from the non edible parts of fishes such as tails, fins and bones.

GENERAL KNOWLEDGE AND APTITUDE

181. Which of the following States has been declared as 'India"s first State to have adopted organic farming?

a) Tripura b) Sikkim c) Arunachal Pradesh d) Nagaland

182. Name the person selected as the 'Glenfiddich's Emerging Artist of the Year' 2019 to represent India?

a) Prabhakar Pachpute b) Dheer Kaku

c) Gipin Varghese d) Raju Baraiya

183. Who has won the Madrid Open Tennis Championship,2019 for Men's Singles Title ?

a) Roger Federer b) Novak Djokovic c) Rafael Nadal d) Tsitsipas

184. Nagaland's traditional Konyak dance sets New Guinness Word Record and how many women participated in this dance ?

a) 2000 b) 3000 c) 5000 d) 6000

185. Pullela Gopichand Badminton Academy collaborated with which of the following Institutes for developing a training module ?

a) IIT-Bombay b) IIT-Kharagpur c) IIT-Kanpur d) IIT-Madras

186. Which is the World's First City With 5G Network Coverage ?

a) Singapore b) Tokyo c) Shanghai d) California

187. Who launched 'Veer Parivar Mobile App' for CRPF personnel killed on duty to help families of martyrs ?

a) Prime Minister b) President c) Home Minister d) Defence Minister

188. Which among the following countries whose National Anthem has only music not words ?

a) India b) Germany c) Bahrain d) Spain

199. Who is the most successful captain in the history of Indian Premier League (IPL) T-20 Cricket ?

a) MS Dhoni b) Virat Kohli

c) Gautham Gambhir d) Rohith Sharma

190. Which Indian armed force has decided to observe 2019 as 'Year of Next of Kin'?

(a) Indian Navy (b) Indian Army c) Indian Air Force d) Indian Coast Guard

191. Select the one which is different from the other three alternatives

a) MNJK b) BCGH c) LMOP d) WXRT

192. If the first and second letters in the word "COMMUNICATIONS' were interchanged, also the third and the fourth letters, the 5th and 6th letters and so on, which letter would be the tenth letter counting from your right ?

a) A b) T c) N d) U

193. A series is given with one term missing. Choose the correct alternative from the given ones that will complete the series

975, 864, 753, 642, _?_

a) 431 b) 314 c) 531 d) 532

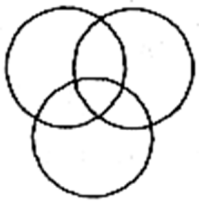
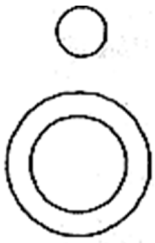
194. If MEKLF is coded as 91782 and LLLJK as 88867, then how can IGHED be coded ?

a) 97854 b) 64521 c) 53410 d) 75632

195. C is the mother of A and B. If D is the husband of B, what is C to D ?

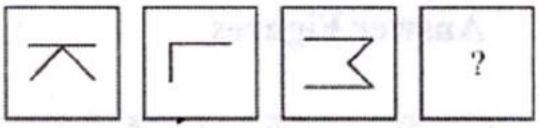
a) Mother – in – law b) Sister c) Mother d) Aunt

196. Which one of the following diagrams represents the correct relationship among 'Judge', 'Thief' and 'Criminal'?

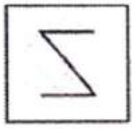
a)  b)  c)  d) 

197. What comes next in the series?

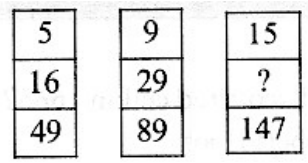
Question Figures:



Answer Figures:

a)  b)  c)  d) 

198. What is the number missing from the third targer?



5	9	15
16	29	?
49	89	147

a) 45 b) 48 c) 51 d) 54

199. BCFE : HILK : : NORQ : ?

a) TXWU b) TXUW c) TUXW d) TUWX

200. Ram and Sham start walking in opposite directions. Ram covers 6 kms and Sham 8 kms. Then Ram turns right and walks 8 kms and Sham turns left and walks 6 kms. How far each is from the starting point?

a) 11 kms b) 8 kms c) 9 kms d) 10 kms